Unit 09: Looping Statements

CONTENTS

Objectives

After studying this unit, you will be able to:

- Explain looping concept in C
- Describe do-while loop
- Describe goto statement

Introduction

Looping must not continue indefinitely as an analogy to real life you would not like to crack the same joke again and again, so a mechanism is required to break out the loop and to allow the executives of the next set of statements.

9.1 Looping

Iteration statements are also known as loops or looping statements because the program execution typically loops through the statement more than once. In this category, C provide the following statement or you call loops.

1. for loop
2. while loop
3. do-while loop

As a result, a general structure for implementing a loop expression has been designed. Which can be better understood by grasping the following elements/parts/components of a loop that regulates the amount of repetitions:

1. Initial Expression(s): Initial expression(s) is usually an assignment expression(s) which initializes the control variable(s) of a loop, as they must be initialized before entering in a loop. The initial expression(s) is executed only once, in the beginning of the loop. But if this expression(s) occurs in

the loop body, control variable(s) would be reassigned to initial values with every loop pass, and the condition expression would never fail.

2. Condition Expression: Conditional expression is typically a relational expression that is set up to terminate the execution of a loop. If the condition expression evaluates, to true i.e. 1, the loop body gets executed, otherwise the loop is terminated. A condition expression may be evaluated before entering in to a loop or before exiting from the loop called as entry-controlled loop and exit controlled loop respectively. In C, the for loop and while loop are entry-controlled loops whereas do while loop is exit-controlled loop.

3. Update Expression(s): The update expression(s) is essentially an increment expression or decrement expression that changes the value(s) of loop variable(s), so that they could come to the boundary values. The update expression(s) normally execute at the end of the loop body. It may appear in the body of loop as it is updating expressions that assign the variable a new updated value every time the loop passes.

4. The Loop Body: The loop body consists of statement(s) that is supposed to be executed again and again as long as the condition expression evaluator to true i.e. 1. In an entry-controlled loop, the condition expression evaluated first and if it evaluates to true, the loop-body is executed and if it evaluate to false, the loop-body is terminated. Whereas, in exit controlled loop, the loop body executed first and then the condition expression are evaluated. It is evaluate to false i.e. o, the loop is terminated, otherwise repeated.

The above mentioned components are the essential component of a statement to be called as a loop statement. Messing any of them may change the basic meaning of a perfect loop. The for, while and do-while statements of C, comprises of all these essential components, hence referred to as loop statements.

for loop

The for loop in C is the simplest, fixed and entry controlled loop. It is simplest as the structure of for loop is divided into two segments i.e. control statement and the body of the loop. All its loop control elements are placed together in the control statement where as body of the loop consists of statements to be executed repeatedly.

It is fixed as number of repetitions is known is advance and can be useful in a situation when you want to do something a fixed number of times. It is an entry controlled loop as the control statement placed before the loop body i.e. condition expression will be evaluated first. The general form of the for loop is:

for(initial expression(s) ; condition expression ; update expression(s)) loop-body;

Example: Consider the following statement:

```
for ( i =1 ; i<= 10; ++i)
printf(" \n Hello World!");
```

where i is an integer variable declared already.

i=1; is an initial expression.

i < = 10; is a conditional expression.

++i; is an update expression.

And the statement

printf("\n Hello World!") ;

is the body of the loop.

When the above statement is encountered during program execution, the following events occur:

1. Initial expression is evaluated first and i will be assigned an initial value 1 i.e. i =1.

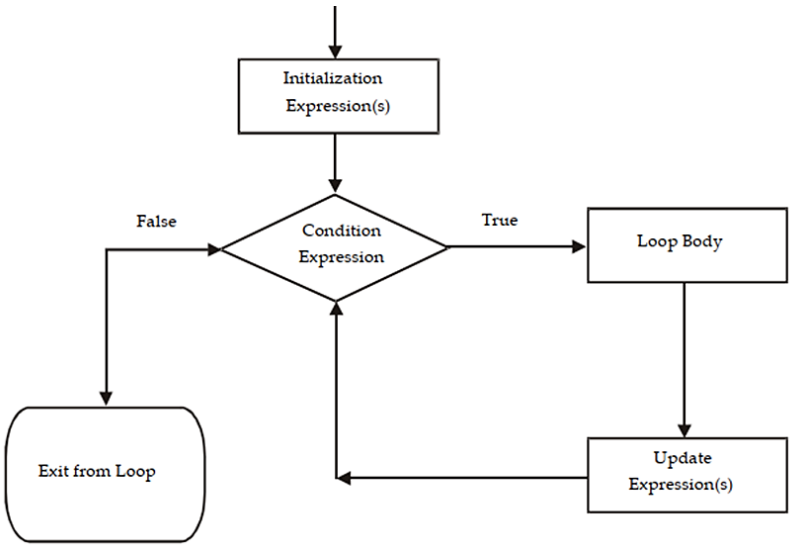
2. Then the condition expression is evaluated i.e. i < =10 and the result will be true as 1 < = 10 is true.

3. Since the condition expression is true, the statement in the loop body is executed i.e. printf("\n Hello World!"); which prints the message Hello World! on the screen.

4. After the execution of the loop body, the update expression i.e. ++i is executed which increment the value of i by 1. In this way after the first execution of the loop the value of I becomes 2 as initially it was 1.

5. After the execution of the update expression the condition expression is again evaluated. If it evaluates to true the sequence is repeated from step no. 3, otherwise the loop terminates.

Also note that the loop body never executes if condition expression is evaluated to false in its first execution. Figure shows the operation of a for loop.



Lab Exercise:

Program :

```
#include<stdio.h>
int main(){

int i,n;
printf("Enter Number");
scanf("%d",&n);
for(i=1;i<=10;i++)
{
   printf("%d*%d=%d\n",i,n,i*n);

}
return 0;

}
```

Output : -

```
Enter Number 5
1*5=5
2*5=10
3*5=15
4*5=20
5*5=25
6*5=30
7*5=35
8*5=40
9*5=45
10*5=50

Process returned 0 (0x0)   execution time : 3.716 s
Press any key to continue.
```

While loop

The while loop, the second type of loop, is an entry controlled loop because it tests the conditions first and only the control enters the loop body if the condition is true.

When the loop's iterations are complete, the control returns to the while statement, which repeats the condition test. If the condition is false the first time, the loop does not iterate and control is passed to the statement after the loop statement. Because we don't know the precise amount of iterations, it's a kind of variable loop. The statements keep repeating themselves until specific conditions are met.

The while loop has the following form:

while (condition expression)

loop body;

Where the loop-body may contains a single statement, a compound statement or an empty statement. The while loop iterates the loop body as long as the specified condition expression evaluates to true.

The while loop doesn't explicitly contains the initialization expression and update expressions of the loop. These two expressions are normally provided by the programmers as the initialization expression(s) should be placed before the loop begins and updation expression(s) should be inside the loop body. By using all these expressions the general farm of while loop may looks like as:

:

initialization expression(s);

while (conditional expression)

{ :

: Loop Body

updation expression;

}

Example:

Consider the following segment of code:

i = 1 ;

```
while ( i < = 10)
{
printf(“\n Hello World!”);
+ + i;
}
```

where i is an integer variable declared already

i = 1; is an initial expression

i < = 10; is a conditional expression

++i ; is an update expression.

And the statements between the { and } forms the body of the loop. But the braces can be discarded, if there is only one statement in the loop body

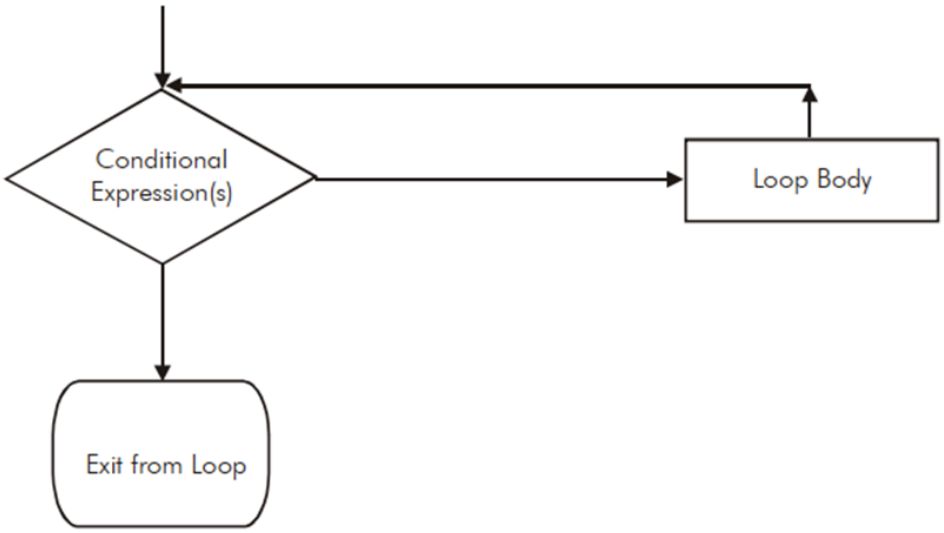
When the program execution readers a while statement, the following events occur:

1. First of all the conditional expression is evaluated i.e. i < 10.

2. The conditional expression is evaluated to true as i was 1 initially and 1 < 10 is true. But if it evaluate to false, the loop will be terminated and the control moves to the first statement following loop body.

3. Since the condition expression is true, the loop body will be executed i.e. the printf statement and the updation expression.

4. With the closing braces (}), it is assumed that the loop is finished and the control moves back to the while statement, which repeats the test again and proceeds accordingly.



Lab Exercise:

Program : -

```
#include<stdio.h>
 int main()
{
int n, reverse=0, rem;
printf("Enter a number: ");
```

```
 scanf("%d", &n);
 while(n!=0)
 {
   rem=n%10;
   reverse=reverse*10+rem;
   n/=10;
 }
 printf("Reversed Number: %d",reverse);
return 0;
}
```

Output : -

```
Enter a number: 1024
Reversed Number: 4201
Process returned 0 (0x0)   execution time : 3.135 s
Press any key to continue.
```

do-while loop

C's third loop statement is the do-while loop, is an exit controlled loop i.e. it tests the conditions after having executed the statement with in loop body. This means unlike the for and while loops, a do while loop always executes at least once. The statement of the do-while loop is as follows:

```
do
{
loop-boody ;
}while (conditional expression) ;
```

The braces { } can be discarded when the loop-boody contains a single statement. The do-while loop iterates the loop body as long as the specified condition is true while testing the condition at the end of the loop each time, rather than at the beginning, as is done by the for and the while loop.

Like while loop, do-while loop also doesn't contain the initialization and updation expression as part of loop statement. However, these expressions can be associated with do-while loop by the programmer according to required logic. Then the new form of do-while loop may looks like as:

```
Initialization expression(s);
do
{
Loop body;
Updating expression;
}while ( conditional expression(s) );
```

Example: Consider the following segment of code :

```
i=1;
```

```
do
{
printf("\n Hello World!");
++i;
} while ( i<=10);
```

where i is an integer variable declared already

i=1; is an initial expression.

i < = 10; is a conditional expression.

++i; is an update expression.

When the program control reaches at a 'do while' loop, the following events occur:

1. The loop body will be executed i.e. the print statement and the updation statement.

2. The conditional expression will be evaluated i.e. i <=10.

3. The conditional expression will evaluates to true as the value as i is 2 this time (initially=1).

4. Since the condition expression is true, the control will move back to execute the loop-body once again.

The output of the above code may looks likes as:

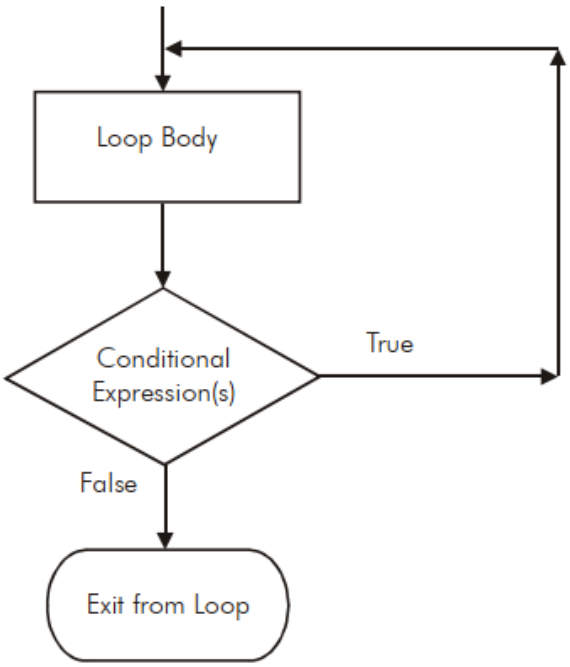
Hello World !

Hello World !

:

Hello World ! (10 times)

Figure demonstrate working of do-while loop



Program to print series of number after enter first number

Lab Exercise:

Program :-

```
#include<stdio.h>
int main(){

int number;
printf("Enter a number:");
scanf("%d",&number);
do{

   printf("Value of number is= %d\n",number);
   ++number;

}while(number<=20);
return 0;

}
```

Output:-

```
Enter a number:10
Value of number is= 10
Value of number is= 11
Value of number is= 12
Value of number is= 13
Value of number is= 14
Value of number is= 15
Value of number is= 16
Value of number is= 17
Value of number is= 18
Value of number is= 19
Value of number is= 20

Process returned 0 (0x0)   execution time : 21.635 s
Press any key to continue.
```

9.2 Jump and Break Statement

What if you need to get out of a loop statement before the test condition fails? The break statement can be used. Break is a statement that is used to end loops or exit from a switch (discussed later). When a break occurs within a C loop, the loop is immediately terminated without verifying the loop condition, and control is passed to the first statement following the loop. It can be used in a while loop, a do-while loop, a for loop, or a switch loop. The break statement is simply written break;

The break statement does not have any operand.

Following C code snippets illustrate use of break statement to exit from various C loops. In each situation, the loop will continue to execute as long as the current value for the integer variable x

does not exceed 10. However, the computation will break out of the loop if a negative value for x is detected.

While loop

```
scanf (“%d”, &x);
while (x <= 10)
{
if (x < 0)
{
printf (“Negative value entered!!\n”);
break;
}
scanf (“%d”, &x);
}
```

do-while loop

```
do
{
scanf (“%d”, &x);
if (x < 0)
{
printf (“Negative value entered”);
break;
}
} while (x < = 10);
```

for loop

```
for (i = 1; x < = 10; ++i)
{
scanf (“%f”, &x);
if (x < 0)
{
printf (“Negative value entered!!”);
break;
}
}
```

When break is used in nested while, do-while, for or switch statements, it will cause a transfer of control out of the immediate enclosing statement, but not out of the outer surrounding statements.

Consider the following code snippet in which a while loop is nested within a for loop.

```
for (i = 0; i < = n; ++i)
```

```
{
while (( c = getchar( ))! = '\n')
{
if (c = '*') break;
}
}
```

9.3 goto Statement

In early programming languages, goto was a common looping idiom for unconditionally branching to one statement from another. For looping immediately, no condition is verified. The usage of goto branching is often discouraged due to the inherent complications connected with it.

C enables the goto statement to branch unconditionally from one point in the programme to another for backward compatibility. After a goto is encountered, a goto statement utilises an identifier called label to specify the statement to which branching will begin. Any valid identifier name must be followed by a colon in a label.A label is placed immediately before the statement where the control is to be transferred.

The general forms of goto and label statements are shown below:

```
goto label;
…..
…..
label: statements;
. . . . . .
statement;
```

Lab Exercise:

```
Program to demonstrate working of goto statement*/
#include<stdio.h>
int main(){

int i;

for(i=0;i<=10;i++)

{
  if(i==5)
  {
    goto STATUS;

  }
  printf("%d\n",i);
```

```
}
STATUS:
   printf("Value of i is = 5 than goto working");
return 0;

}
```

Output :-

```
0
1
2
3
4
Value of i is = 5 than goto working
Process returned 0 (0x0)   execution time : 0.013 s
Press any key to continue.
```

Summary

- Most of the programs require a statement or set of statements to be executed multiple times or not to execute at all, depending on the circumstances.
- The statement by which we can control the flow of the program execution is called as control flow statement or program control statement.
- In the sequence construct, as the name implies, statements are executed sequentially i.e. one after the other.
- In selection construct, the execution of statements depends upon a condition test.
- The iteration constructs are an efficient method of handling a series of statements that must be repeated a variable number of times.
- If multiple statements are to be executed than they must be placed with in a pair of braces.
- A simple if or if else construct may be placed with in another if or if-else construct.
- The switch statement is another convenient tool provided by C to handle the situations in which multiple decisions to be made based on an expression that can have multiple values.
- The for loop in C is the simplest, fixed and entry controlled loop. An infinite for loop can be created by skipping the conditional expression.
- A conditional expression cannot have multiple expression like initialization and updation expression, but it may contain several conditions linked together using logical operators.
- The second type of loop, the while loop is an entry controlled loop as it tests the conditions first and if the condition is true, then only the control will enter into the loop body. An empty loop can also be configured using while statement and could used as a time delay loop.
- C's third loop statement is the do while loop, is an exit controlled loop i.e. it tests the conditions after having executed the statement with in loop body. Unlike the for and while loops, a do while loop always executes at least once.
- The break statement is used in a program to skip the particular part of program code. The jump statement continue is the compliment of the break statement.

Keywords

Conditional statement: A statement that evaluates to either true or false.

Continue statement: The statement that ignores execution of further statements and forces the loop to evaluate the loop condition once again.

Default statement: An optional statement in a switch that is executed if none of the conditions evaluates to true.

Switch statement: A multi-selection statement that branches to that statement whose specified condition evaluates to true.

Control Statements: The statements that allow programmers to alter the sequential flow of execution of the program and control the flow are called control statements.

For Loop: A for loop allows execution of a statement (or a block of statements) repeatedly a number of times.

While Loop: In case the number of times a statement is to be executed is not known in advance, while loop is used.

goto statement : The goto statement is known as jump statement in C.

Self Assessment

1. What are the components of design structure of C?

A. Documentation

B. Link Section

C. Definition Section

D. Above all

2. Every C program must have _____ function.

A. Printf ( )

B. Scanf ( )

C. main ( )

D. none of above

3. What are the Advantages of Top-Down Design?

A. Important modules are designed first

B. Testing and debugging is easier and fast

C. Project implementation is easy

D. All of the above

4. How many type of conversion are there in c?

A. 3

B. 2

C. 1

D. 4

5. Which conversion also called Automatic Type Conversion?

A. Explicit Type Conversion
B. Implicit Type Conversion
C. All of above
D. None of the above

6. Which type of conversion is NOT accepted?

A. From char to int
B. From float to char pointer
C. From negative int to char
D. From double to char

7. Type modifiers in C are

A. Int, main
B. Sub program
C. Short, long, signed and unsigned
D. All of above

8. Which one is decision making statement in C ?

A. If statement
B. While loop
C. Switch
D. For loop

9. What will be output for the following code?

```
#include <stdio.h>
  void main()
  {
     int x = 5;
     if (true);
        printf("hello");
  }
```

A. It will display hello
B. It will throw an error
C. Nothing will be displayed
D. Compiler dependent

10. Which of the following is an invalid if-else statement?

A. if (if (a == 1)){}
B. if (func1 (a)){}
C. if (a){}
D. if ((char) a){}

11. The label in goto statement is same like

A. Case in switch statement

B. Initialization in for loop

C. Continuation condition in for loop

D. All of them

12. Goto statement is also known as _____

A. If statement

B. Jumping statement

C. Loop statement

D. Above all

13. Choose a right C Statement.

A. Loops or Repetition block executes a group of statements repeatedly.

B. Loop is usually executed as long as a condition is met.

C. Loops usually take advantage of Loop Counter

D. All the above.

14. The break statement is used to exit from:

A. a DO loop.

B. a FOR loop.

C. a SWITCH statement.

D. all of above.

15. Which keyword is used to come out of a loop only for that iteration?

A. break

B. continue

C. return

D. none of the mentioned

Answers for Self Assessment

1.	D	2.	C	3.	D	4.	D	5.	B
6.	B	7.	C	8.	A	9.	B	10.	A
11.	A	12.	B	13.	D	14.	D	15.	B

Review Questions

1. Write a program using if-else statement.

2. Explain nested-if statement with example.
3. What do you mean by switch statement? How it used
4. A five-digit number is entered through the keyboard. Write a program to obtain the reversed number and to determine whether the original and reversed numbers are equal or not.
5. Write a program to check whether a triangle is valid or not, when the three angles of the triangle are entered through the keyboard. A triangle is valid if the sum of all the three angles is equal to 180 degrees.
6. Given the length and breadth of a rectangle, write a program to find whether the area of the rectangle is greater than its perimeter. For example, the area of the rectangle with length = 5 and breadth = 4 is greater than its perimeter.
7. What is the use of if-else statement?
8. Define selection in c programming.
9. Write a program in C to enter five integer values as age of five boys and calculate the average age of all the boys.
10. Write a program to calculate the area of a square. All values enter with the help of keyboard.
11. What do you mean by looping?
12. Describe for loop with the help of suitable example.
13. Differentiate while loop and do-while loop.
14. What is the advantage of break statement in while loop?
15. Write a program to find the factorial value of any number entered through the keyboard.
16. Write a program to print all the ASCII values and their equivalent characters using a while loop. The ASCII values vary from 0 to 255.

Further Readings

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